

Autonomous UAV SITL Validation with Slung Payload

PX4 + Gazebo Classic + ROS 2 offboard pipeline, payload-link debugging, hover validation, and 8-shaped trajectory results

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Repository	https://github.com/paneendrakumar0/uav-autonomous-telemetry
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Primary milestone	Native ball-joint slung-payload Figure-8 completed

This report summarizes the development and validation of an autonomous multirotor simulation pipeline for trajectory tracking with a slung payload. It is written as a concise technical record of what was built, what failed, what was corrected, and what evidence is now available for review.

Executive Summary

The project now has a working PX4 SITL and ROS 2 offboard-control pipeline that can command an Iris-derived quadrotor through a continuous 8-shaped trajectory. The final validated model is a native Gazebo Classic SDF named **iris_depth_payload**, derived from the standard Iris model and extended with a physical internal ball joint from **base_link** to **slung_payload**.

The final slung-payload Figure-8 run lasted **150.91 s**. After the first 25 s transient window, the UAV achieved a mean 3D tracking error of **0.462 m** and RMS error of **0.493 m**, while maintaining mean altitude at **-4.996 m NED** for a commanded altitude of -5 m.

A major engineering finding was that direct joints across nested Gazebo models caused the vehicle to accept PX4 offboard commands but remain near the ground. The issue was isolated through controlled hover tests and solved by moving the payload link into the same native vehicle model. Payload collision was disabled in the current flight baseline to avoid ground-contact solver locking during spawn; payload mass, visual cable, visual payload, and pose logging remain active.

System Architecture

Layer	Implementation
Flight stack	PX4 Autopilot SITL
Simulator	Gazebo Classic 11
Middleware	Micro XRCE-DDS Agent
Autonomy/control	ROS 2 Humble offboard position/velocity setpoint nodes
Vehicle baseline	Iris with depth camera
Final payload vehicle	Native Iris-derived iris_depth_payload SDF
Payload model	0.05 kg spherical payload, 1 m visual cable, internal ball joint
Primary outputs	CSV telemetry, trajectory plots, payload swing metrics, Gazebo screenshots

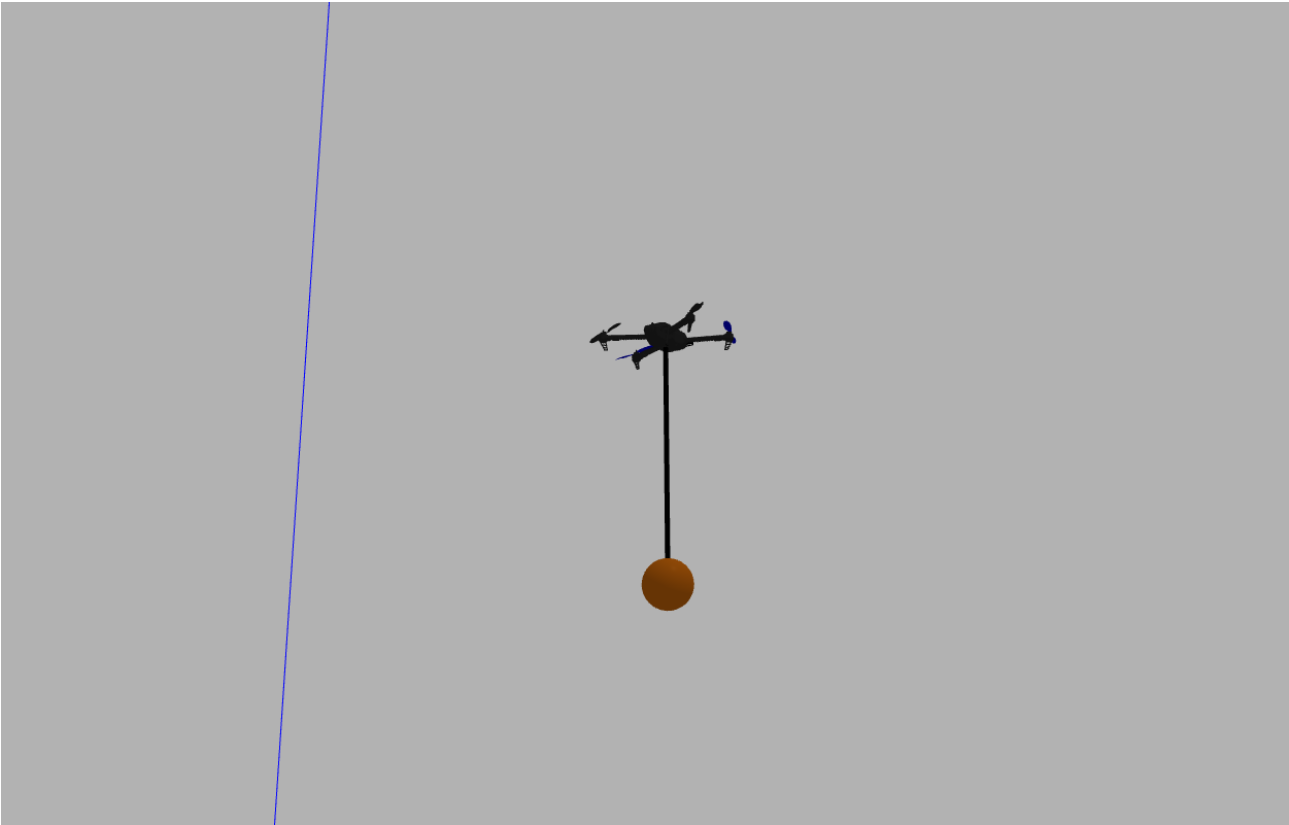


Figure 1. Final native ball-joint slung-payload model hovering in Gazebo: UAV body, cable, and orange payload are visible.

Development and Validation Timeline

Stage	Outcome	Evidence
No-payload Figure-8	Tuned trajectory tracking achieved mean post-takeoff error 0.415 m.	fig8_metrics_tuned_2026-06-04
No-payload hover control check	Hover controller validated independently: mean error 0.067 m.	hover_control_check_2026-06-05
Initial payload attempts	Payload SDF/link variants armed but did not climb; single-link mean error 0.660 m.	payload_500m_singlelink_2026-06-05
Joint isolation	Free nested payload hovered; fixed/ball cross-model joints failed. Payload hover was isolated free.	payload_isolated_free_2026-06-05
Native payload link fix	Native internal ball joint hovered with mean error 0.061 m.	payload_hover_native_ball_nocollision_2026-06-05
Final payload Figure-8	Sustained 8-shaped trajectory with mean post-25s error 0.462 m.	payload_figure8_native_ball_2026-06-07

Baseline Figure-8 Without Payload

The first successful autonomy milestone was a continuous Figure-8 trajectory without payload. This validated the PX4 offboard message path, the Micro XRCE-DDS bridge, and the ROS 2 trajectory generation and telemetry logger. The tuned run used a slower angular rate to obtain a cleaner steady-state response.

Metric	Value
Post-takeoff mean error	0.415 m
Post-takeoff RMS error	0.442 m
Post-takeoff max error	0.660 m
Duration	86.09 s

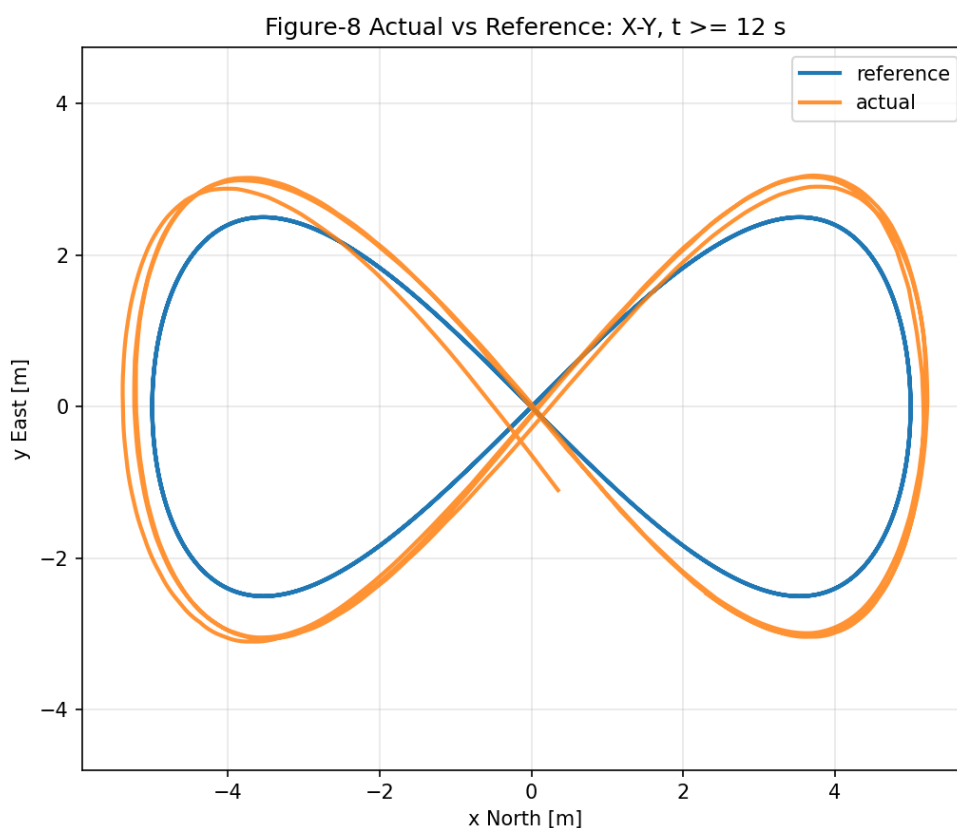


Figure 2. Tuned no-payload Figure-8 steady-state XY tracking.

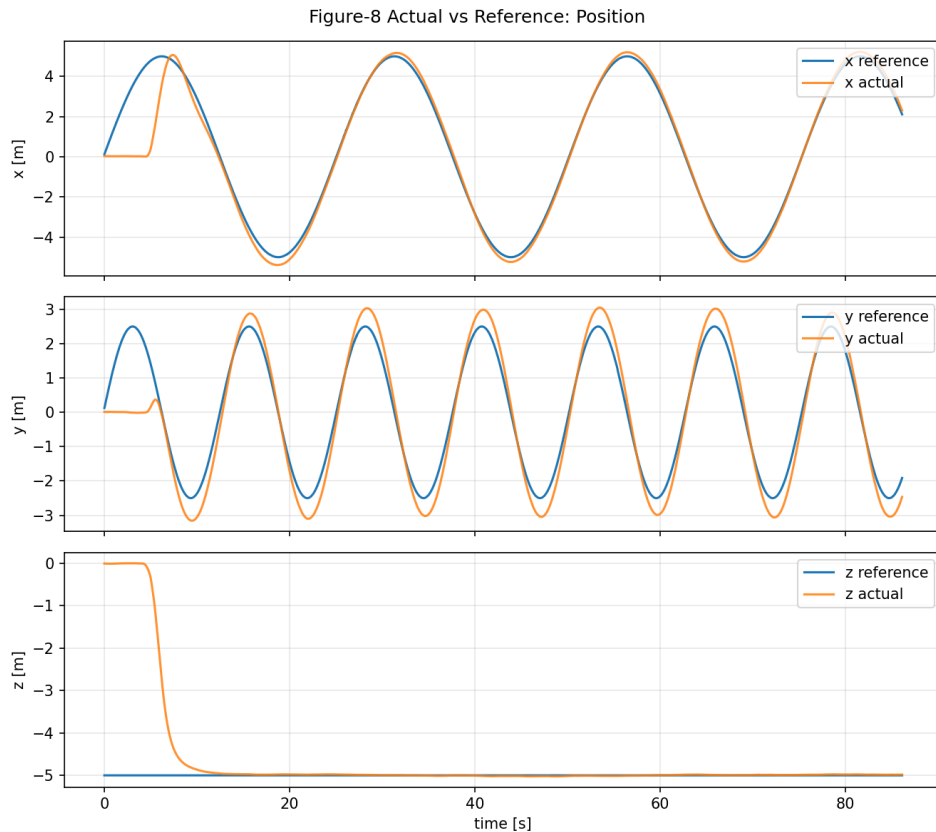


Figure 3. Tuned no-payload Figure-8 XYZ coordinates versus time.

Payload Link Debugging

The most important failure mode was not in the ROS 2 controller. PX4 accepted offboard mode and arming commands, and the vehicle reported takeoff, but the Gazebo model remained near the ground when the payload was attached through several direct joint variants. This was reproduced across ball, fixed, and simplified single-link configurations.

The debugging strategy was to simplify one variable at a time. A no-payload control test confirmed the hover controller. A free nested payload model confirmed that merely having a payload in the world did not break flight. A fixed joint to the payload reproduced the failure. Finally, rebuilding the vehicle as a native Iris-derived SDF and moving the payload link inside the same model solved the climb problem once payload collision was removed.

Case	Final/mean result	Interpretation
No-payload hover	Mean error 0.067 m	Controller and PX4 bridge are valid.
Single-link payload failure	Mean error 5.008 m	Payload joint/model setup was blocking flight.
Native ball-joint hover	Mean error 0.061 m	Native internal payload link solves hover.



Figure 4. Earlier slung-payload visual/debug model; the physical joint formulation still prevented climb.

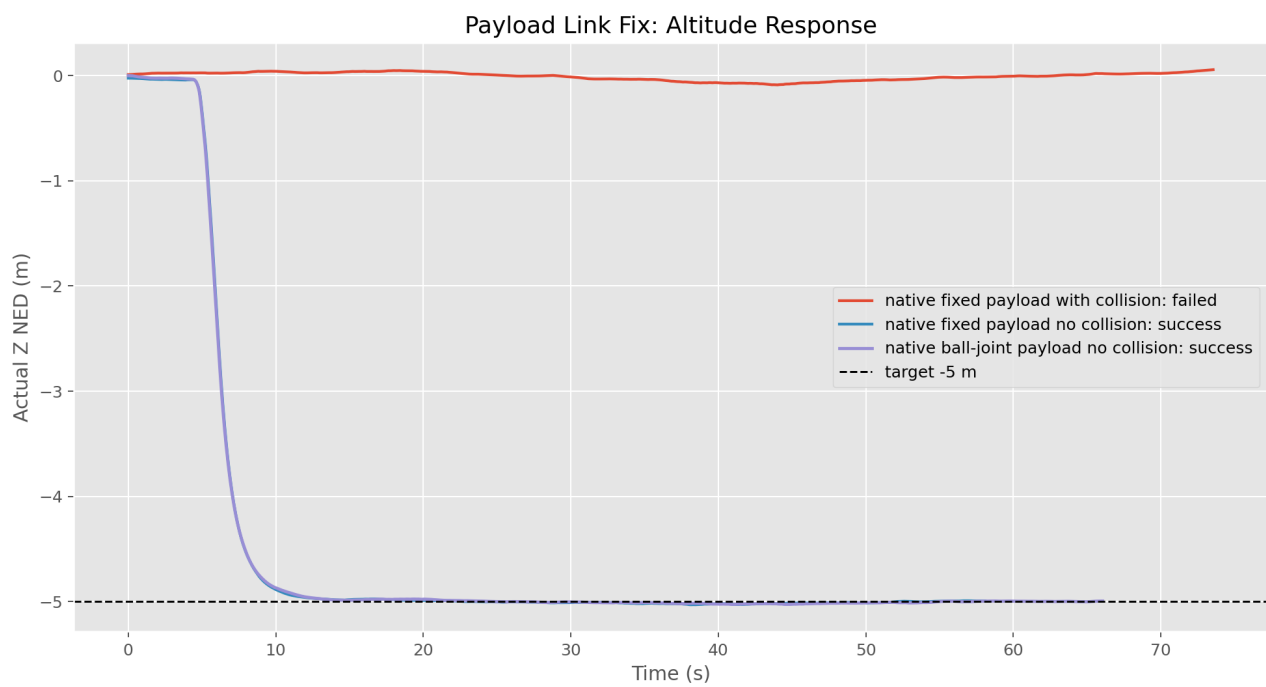


Figure 5. Payload link fix comparison: collision/contact and joint-frame choices determined whether the vehicle could climb.

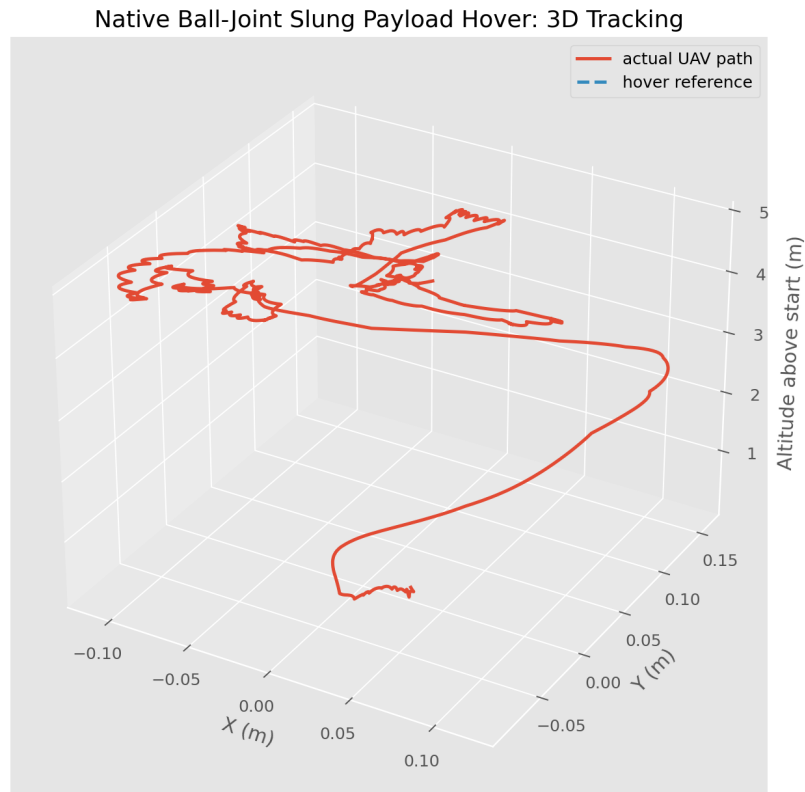


Figure 6. Native ball-joint payload hover: 3D tracking around the -5 m NED target.

Final Payload Figure-8 Validation

The final validation directly addresses the requested 8-shaped autonomous circuit. The native slung-payload model was commanded with PX4 offboard position and velocity setpoints using a lemniscate trajectory with 5 m amplitude and 0.25 rad/s angular rate at -5 m NED altitude.

Metric	Value
Duration	150.91 s
Samples	7545
Post-25s mean 3D error	0.462 m
Post-25s RMS 3D error	0.493 m
Post-25s max 3D error	0.796 m
Mean post-25s altitude	-4.996 m NED
Final altitude	-4.987 m NED
Actual X range	-5.489 m to 5.291 m
Actual Y range	-3.158 m to 3.083 m

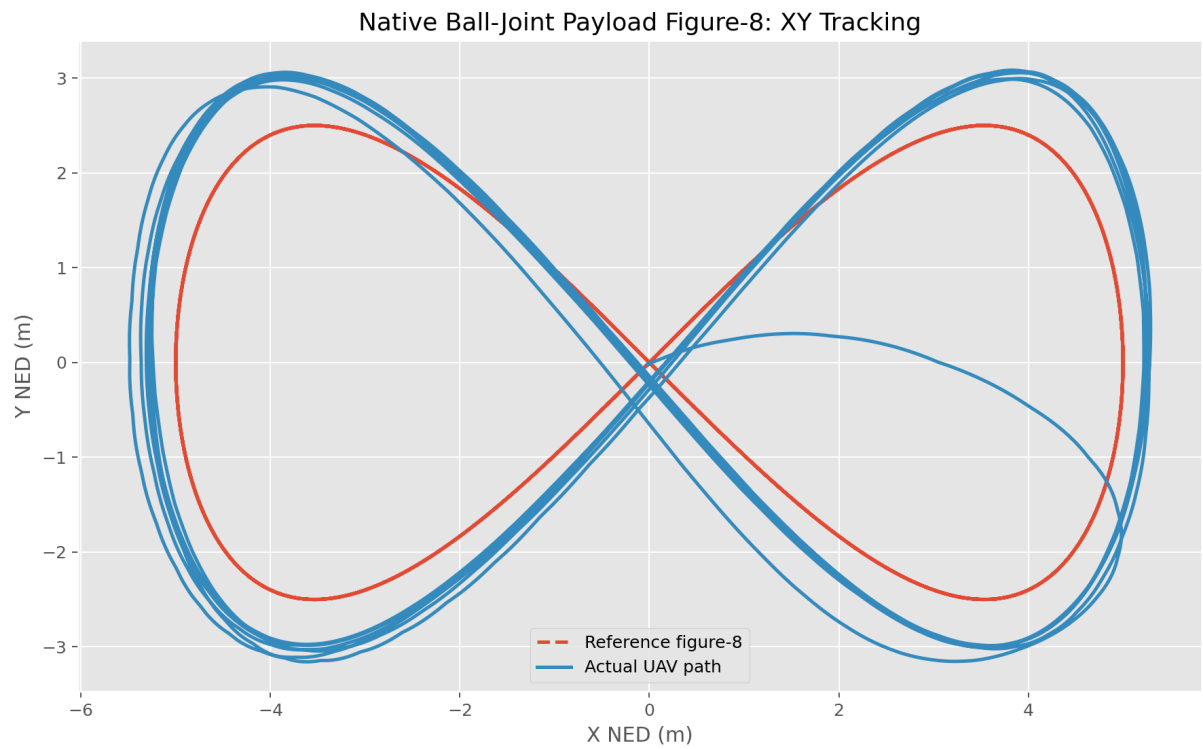


Figure 7. Full native slung-payload Figure-8 XY tracking.

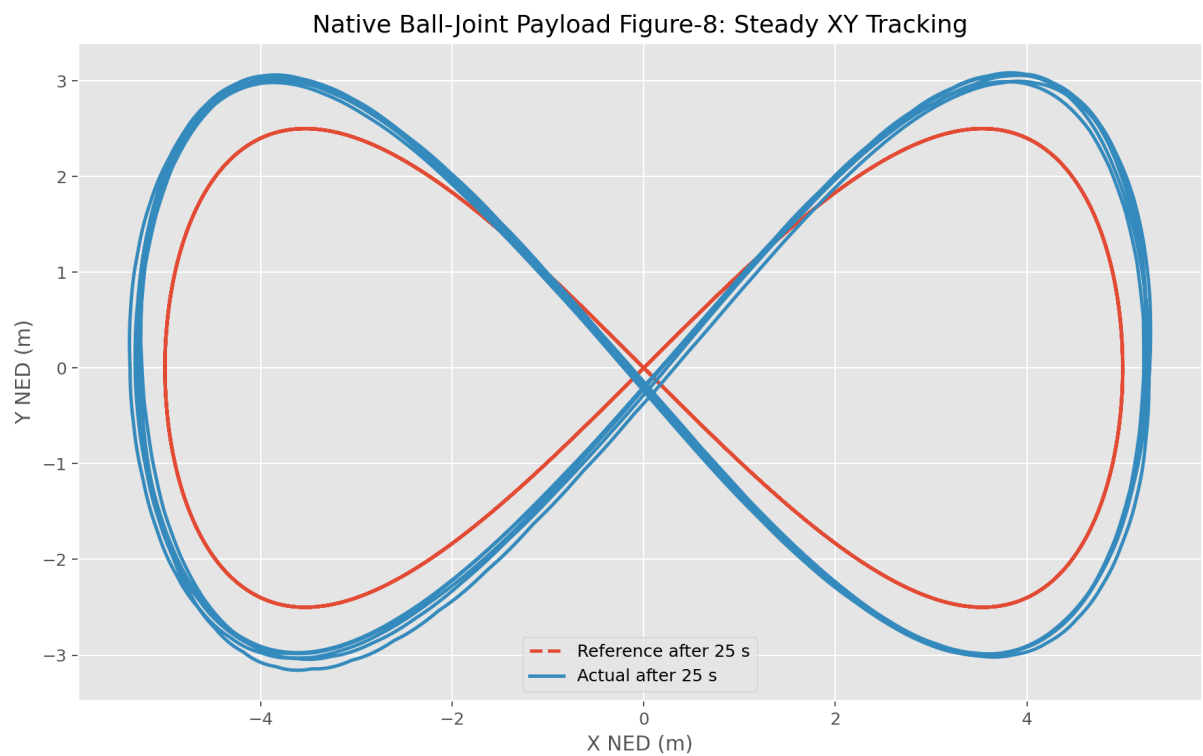


Figure 8. Steady-state native slung-payload Figure-8 XY tracking after the takeoff transient.

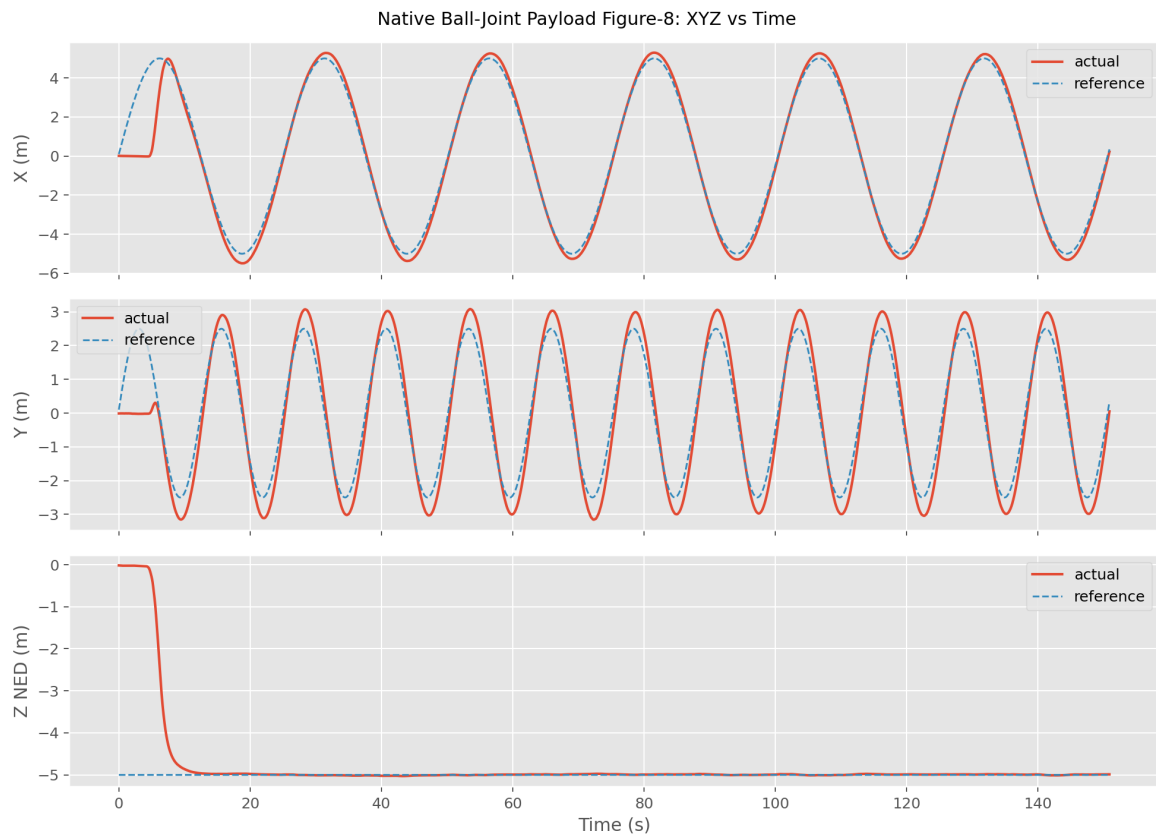


Figure 9. Native slung-payload Figure-8 coordinates versus time.

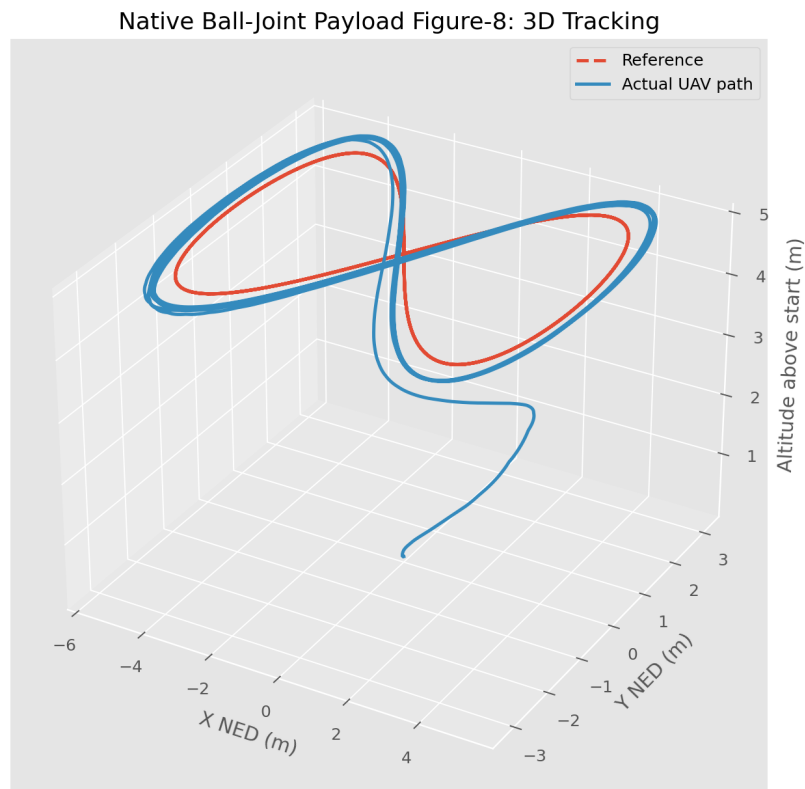


Figure 10. Native slung-payload Figure-8 3D trajectory.

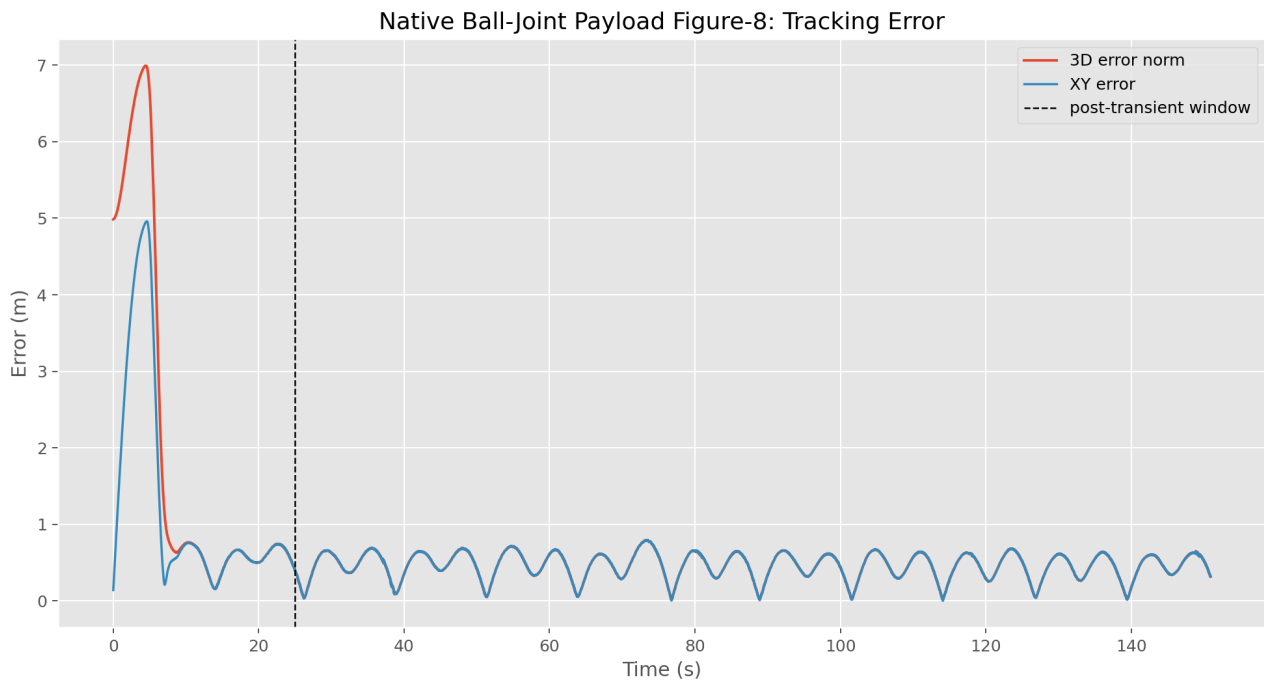


Figure 11. Native slung-payload Figure-8 tracking error.

Payload Motion Diagnostics

The payload pose sniffer records relative motion between the UAV body and the slung payload. These values are useful for qualitative and comparative diagnosis, although the cable-angle estimate is currently sensitive to the Gazebo pose frame convention. Therefore, the payload swing values should be interpreted as diagnostic estimates rather than final physical measurements.

Metric	Value
Payload samples	37644
Mean lateral swing	5.149 m
Maximum lateral swing	10.646 m
Mean cable angle estimate	75.289 deg
Maximum cable angle estimate	86.271 deg

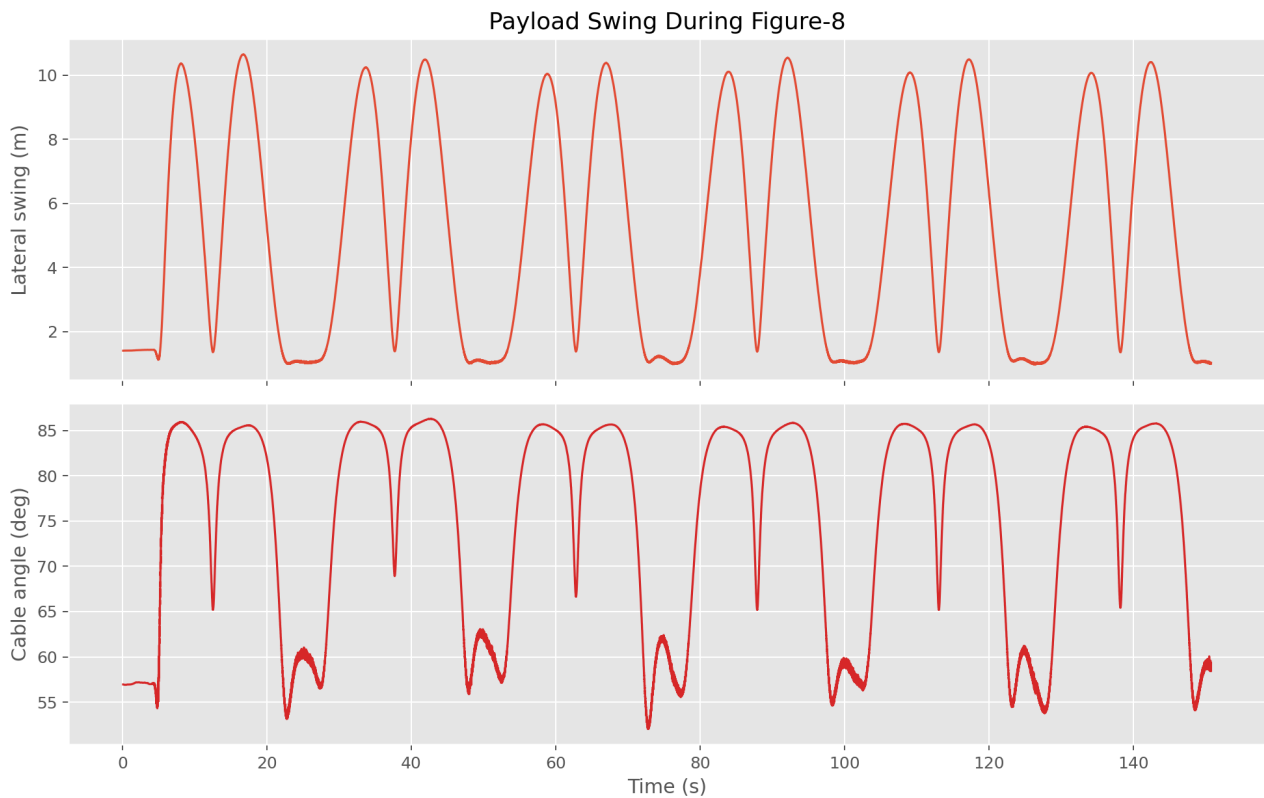


Figure 12. Payload swing diagnostics during the native slung-payload Figure-8 run.

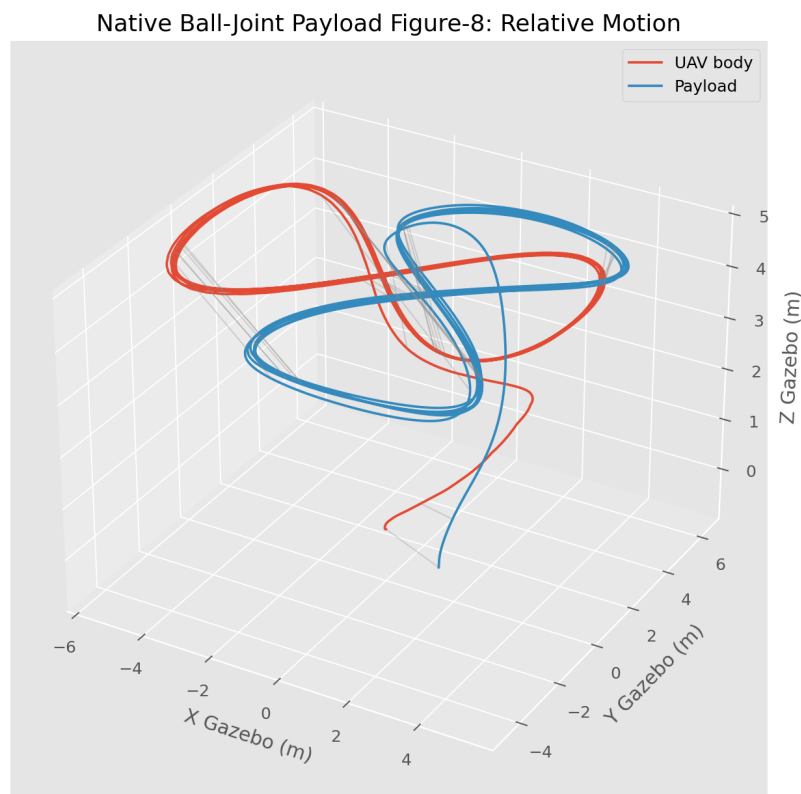


Figure 13. UAV and payload relative motion from Gazebo pose logging.

Main Findings

Finding	Evidence
PX4/ROS 2 offboard path is functional.	No-payload hover and no-payload Figure-8 both track correctly.
The original payload failure was not controller-related.	PX4 armed and reported takeoff while jointed payload SDF variants stayed near ground.

Finding	Evidence
Cross-model payload joints are unsafe for this Gazebo class payload.	Native class payload flew; cross-model fixed/ball joints failed.
Native model integration fixes the link problem.	Internal base_link to slung_payload ball joint hovers and completes Figure-8.
Payload collision must remain disabled in the current baseline.	On baseline contact at spawn caused solver locking and prevented climb.
The 8-shaped circuit requested by the professor is final complete.	Final complete 91 s, mean post-25s error 0.462 m.

Limitations and Next Work

The present result is a strong simulation milestone, but it is still a SITL baseline rather than a final physical-payload controller. Payload collision is disabled to prevent ground-contact locking, so contact-rich payload behavior is not yet represented. The payload swing logger also needs frame-convention refinement before cable-angle values are treated as high-fidelity physical measurements.

The next technical steps are: refine the payload pose/swing estimator, add a controlled payload perturbation test, evaluate trajectory tracking under different amplitudes and rates, and then consider a geometric controller or attitude-reference controller for more aggressive load transport.

Repository and Reproducibility

All source files, ROS 2 nodes, PX4 integration files, CSV logs, generated plots, and report artifacts are available in the GitHub repository: <https://github.com/paneendrakumar0/uav-autonomous-telemetry>. The most relevant final artifacts are stored under **reports/payload_figure8_native_ball_2026-06-07/** and **reports/payload_hover_native_ball_nocollision_2026-06-05/**.

Artifact	Location
Final Figure-8 report	reports/payload_figure8_native_ball_2026-06-07/PAYLOAD_FIGURE8_VALIDATION_REPORT.md
Final Figure-8 CSV	reports/payload_figure8_native_ball_2026-06-07/figure8_tracking_metrics.csv
Payload swing CSV	reports/payload_figure8_native_ball_2026-06-07/payload_swing_metrics.csv
Native payload SDF	px4_payload_integration/.../models/iris_depth_payload/iris_depth_payload.sdf
This PDF	reports/final_research_report_2026-06-07/uav_slung_payload_sitl_research_report_2026-06-07.pdf